

Unit 4, Lesson 9: Absolute Value – Part 2

Benchmark

MA.6.NSO.1.3 Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers.

Learning Target

- ☐ I can find the absolute value of a value and interpret it within the context.

4.9.1 Warm-Up: Absolute Value

1. Choose any value 1 through 9, along with its opposite, to complete the statement below:

_____ and _____ are _____ units away from _____ on the number line.

2. Switch your answer with a partner to check.

3. Ask your partner their strategy for finding their answer and write their strategy below.

4.9.2 Guided Practice: Absolute Value and Opposites in the Real World

1.

A tiger was rescued by The Brevard Zoo in Melbourne, Florida. The zoo veterinarian examined the tiger and determined the animal’s weight to be 29.6 pounds below average.

a.

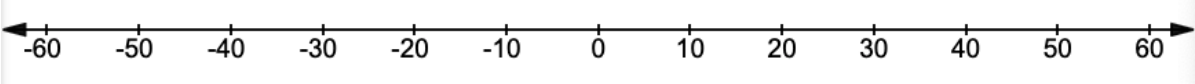
What value represents this situation?

b.

What is the absolute value of the value that represents the tiger’s weight loss?

c.

Model the tiger’s weight loss on the number line.



d.

What does 0 represent in this situation?

e.

If the tiger’s weight had been 29.6 pounds above average instead of below, what would be the same about this situation?
2.

Simone is doing research on different rides at the Strawberry Festival. She attaches an accelerator to a bumper car to measure the velocity of her car. She recorded the velocities, in miles per hour (mph), in the table.

Speed is the absolute value of velocity.

Time (seconds)	Velocity (mph)	Time (seconds)	Velocity (mph)	Time (seconds)	Velocity (mph)
5	−3.1	20	−2.4	35	−5.2
10	−2.7	25	−6.3	40	3.1
15	4.8	30	2.7	45	−7.8

- a.

Determine when the bumper car is at the same speed. List the times and the speed of the car at those times.
- b.

At what time is the bumper car traveling at the slowest speed?
- c.

At what time is the bumper car traveling at the fastest speed?
3.

Before Bryan goes snow skiing, he asks a friend about the weather. The friend says that the temperature is 5°C away from 0.

Select all the temperatures that are possible.

☐ -10°C

☐ -5°C

☐ 0°C

☐ 5°C

☐ 10°C
4.

A part of the city of New Orleans is 19 feet below sea level. We can use “ -19 feet” to describe its elevation, and “ $|-19|$ feet” to describe its vertical distance from sea level. In the context of elevation, what do each of the following numbers describe?

Value	Meaning in Context
25 feet	
$ 25 $ feet	
-8 feet	
$ -8 $ feet	

4.9.3 Collaboration: Absolute Value and Opposites in the Real World

1. Anthony owes his friend \$2.80.

a. Sketch a number line that represents this situation.

b. Represent this situation as a rational number.

c. What is the absolute value of the rational number?

d. What does 0 represent in this situation?
2. With your partner, describe the meaning of each value in the context of temperature, where the freezing point is 0°C.

Temperature	Interpretation
1°C	
−4°C	
12 °C	
−7 °C	

1. The absolute value of the elevation of Enterprise, Florida, is 19.7 feet. In relation to sea level, write the two values that could represent the elevation of Enterprise, Florida.
2. Jasmine received a notice that her bank account was overdrawn, showing a balance of $-\$20.48$. What does the absolute value of her balance mean in this situation?

4.9.5 Wrap-Up: Ticket Out the Door

1. The number -220 represents a withdrawal in dollars from a checking account, and 220 represents a deposit in dollars to the same checking account. What do the absolute values of these values mean in terms of this context?
2. A bank account has an account balance of \$150. For each description, check the value(s) of each expression that correctly matches.

Description	12	$ -12 $	$ 12 $	-12
a withdrawal of \$12	☐	☐	☐	☐
a deposit of \$12	☐	☐	☐	☐
a change of \$12 above the balance	☐	☐	☐	☐
a change of \$12 below the balance	☐	☐	☐	☐

